

Cold Diffusion: Do Diffusion Models Actually Need Noise?

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1. Introduction

Diffusion models are the framework behind DDPM and Stable Diffusion and are built on the assumption that the degradation process is Gaussian noise. Cold Diffusion (Bansal et al., NeurIPS 2023) asks whether we actually need that, replacing Gaussian noise with deterministic degradation such as blur, masking, and downsampling. The paper's central contributions are (1) a unified diffusion framework that generalizes diffusion beyond Gaussian noise to arbitrary deterministic image transforms, and (2) Algorithm 2, an iterative sampling procedure that remains stable to model errors across deblurring and GMM-based image generation.

2. Chosen Result

We reproduce Table 1 of the paper containing quantitative deblurring metrics (FID, SSIM, RMSE) for the Degraded baseline, the one-shot direct reconstruction, Algorithm 1's sampling, and Algorithm 2's iterative sampling, on MNIST (28×28 grayscale handwritten digits) and CIFAR-10 (32×32 RGB natural photographs).

Dataset	Degraded			Sampled			Direct		
	FID	SSIM	RMSE	FID	SSIM	RMSE	FID	SSIM	RMSE
MNIST	438.59	0.287	0.287	4.69	0.718	0.154	5.10	0.757	0.142
CIFAR-10	298.60	0.315	0.136	80.08	0.773	0.075	83.69	0.775	0.071
CelebA	382.81	0.254	0.193	26.14	0.568	0.093	36.37	0.607	0.083

The paper's headline empirical claim is that Algorithm 2 produces better FID than Direct, even when pixel-focused metrics (SSIM, RMSE) are slightly worse. This is not obvious because, without noise to denoise away, we don't know for sure that iterative refinement should help at all. This result is the central evidence for the paper's thesis, which is that iterative diffusion is a real generative framework rather than a noise-specific phenomenon.

3. Methodology

We rebuilt the paper's pipeline from scratch, scaling each component down to fit our compute budget: a smaller U-Net, two datasets instead of the paper's four, and 100k training steps versus the paper's 700k. We generated degraded image trajectories using MNIST and CIFAR-10, and then fed these trajectories into the model for training it to reverse image degradation, and for further evaluation. We used a DDPM U-Net that receives the degraded image and timestep and passes it through an encoder that scales the image down, an attention bottleneck, and then a decoder that scales the image up with skip connections from the encoder. To produce the "noisy" input images to this model, we used a degradation schedule that performed a Gaussian blur at each timestep by applying a 2D Gaussian kernel on the image. We would then assign each dataset a degradation 'blur routine' which would affect the standard deviation of the Gaussian blurring across timesteps, where MNIST had constant standard deviations while CIFAR-10 used linearly increasing standard deviations.

To compare our model to the paper's data, we used the 3 metrics from the paper: FID (distance between deep feature distributions of real and generated images), SSIM (perceived visual similarity

between real and reconstructed images), and RMSE (pixelwise error). After training our model, we used these metrics to compare original images to the model’s reconstruction of their blurred versions.

4. Results & Analysis

Our results show the paper’s trend on every metric (note that our Alg 2 column corresponds to the paper’s Sampled column). On CIFAR-10, Algorithm 2 beats Direct on FID (55.75 vs 58.33; paper: 80.08 vs 83.69), while Direct beats Algorithm 2 on SSIM (0.6736 vs 0.6634; paper: 0.775 vs 0.773) and on RMSE (0.0810 vs 0.0826; paper: 0.071 vs 0.075). On MNIST, the same three orderings hold: Algorithm 2 wins FID (22.69 vs 27.04; paper: 4.69 vs 5.10), and Direct wins SSIM (0.8458 vs 0.7537; paper: 0.757 vs 0.718) and RMSE (0.1358 vs 0.1780; paper: 0.142 vs 0.154).

The paper claims Algorithm 2 is necessary for non-noise degradations but gives no quantitative deblurring comparison, so we included Algorithm 1 in our table. Algorithm 1 underperforms Algorithm 2 on every metric in our experiments, confirming that Algorithm 2’s correction term is required for non-noise degradations.

Dataset	Degraded			Direct			Alg 1			Alg 2		
	FID	SSIM	RMSE	FID	SSIM	RMSE	FID	SSIM	RMSE	FID	SSIM	RMSE
MNIST	398.14	0.0141	0.2866	27.04	0.8458	0.1358	41.16	0.6234	0.2374	22.69	0.7537	0.1780
CIFAR-10	33.65	0.3562	0.1359	58.33	0.6736	0.0810	96.42	0.5188	0.1205	55.75	0.6634	0.0826

Our results did have a few differences. MNIST degraded SSIM is 20× lower than the paper’s (0.0141 vs 0.287). This gap shows up at the degraded baseline itself, which points to a difference in SSIM implementation (likely a difference in window size, data range, or channel handling between our torchmetrics call vs. the paper’s) rather than a difference in reconstruction quality. The downstream Direct and Alg 2 SSIMs inherit this offset, but relative orderings between methods are reproduced. Absolute FID magnitudes diverge in opposite directions across datasets: our MNIST FIDs are roughly 5× higher than the paper’s (22.69 / 27.04 vs 4.69 / 5.10), consistent with our reduced training budget on a smaller U-Net. CIFAR-10 FIDs come in below the paper’s (55.75 / 58.33 vs 80.08 / 83.69), which makes sense as we start from a slightly less degraded image (overall easier starting point for the model and therefore better performance).

To further test the paper’s framework, we implemented two custom degradation schedules beyond the Gaussian blur: the first removes a random color at every timestep, and the second applies a Gaussian blur weighted by the brightness of the center pixel. Although the reverse-degraded images had some resemblance to the original structure, evident by their sizable SSIM score, these degradation schedules greatly underperformed compared to Gaussian blurring. Still, we were able to replicate the improvement from Algorithm 2 sampling, with an improvement on all metrics. Furthermore, the nature of color removal being uniform across the image yet lossy, while Gaussian blur varies heavily across an image while preserving more data, lets us infer that it is more important for our degradation to preserve structure rather than raw pixel values.

Overall, Cold Diffusion’s main claim holds in our implementation. Iterative sampling under deterministic degradation produces better outputs than one-shot reconstruction, at a small cost in pixel accuracy (slightly lower SSIM, slightly higher RMSE). Even with 1/7 of the paper’s training and a smaller model, the difference between Direct and Algorithm 2 reproduces on CIFAR-10 and MNIST, and the Algorithm-1-vs-2 gap is clear on both datasets. This is evidence that diffusion’s iterative refinement carries generative value independent of noise.

Method	Color			Focused Blur		
	FID	SSIM	RMSE	FID	SSIM	RMSE
Degraded	398.14	0.0141	0.2866	398.14	0.0141	0.2866
Direct	314.47	0.0276	0.3457	385.89	-0.0094	0.4333
Alg 1	358.55	0.0291	0.5448	345.62	0.0389	0.3101
Alg 2	302.72	0.2732	0.253	347.28	0.0155	0.2849

5. Reflection

Throughout this project, we learned that small per-step prediction errors compound into measurable FID differences, which is exactly what motivates the Algorithm 2 iteration. Secondly, we learned that degradation starting values greatly impact the resulting reconstruction metrics. Additionally, we learned that SSIM/FID implementations are not directly comparable across codebases and can cause differences as a result. Throughout the project, we all gained significant experience with the full pipeline of model development and training, data arrangement, and especially model evaluation. Also, figuring out why the results differ from the original paper by comparing our implementation step by step to the paper's original was insightful. Furthermore, dealing with the limited compute issues (training notebooks crashed a couple of times due to RAM overload) taught us how to engineer around compute limitations. The biggest limitation was training on the CelebA dataset, which the paper used for the generation experiments. In our setup, training only reached about 1500 of 90,000 steps in 4 hours. Given more time and compute, we would scale the U-Net to the paper's width, train for more steps to get better numbers, and train on CelebA so we could actually run the GMM-based generation experiment.

6. References

1. A. Bansal et al. Cold Diffusion: Inverting Arbitrary Image Transforms Without Noise. arXiv:2208.09392; NeurIPS 2023.
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3. A. Krizhevsky. Learning Multiple Layers of Features from Tiny Images. Technical Report, University of Toronto, 2009. (CIFAR-10.)
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Github repo link: <https://github.com/NickyJosephson/cs4782-final-project>